

ICML 2026

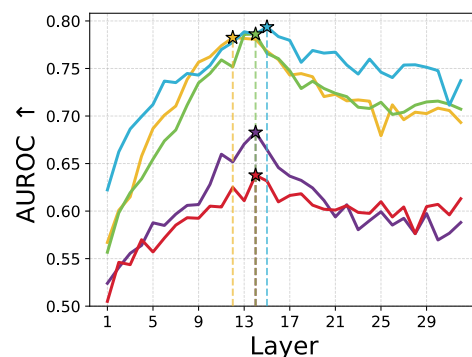
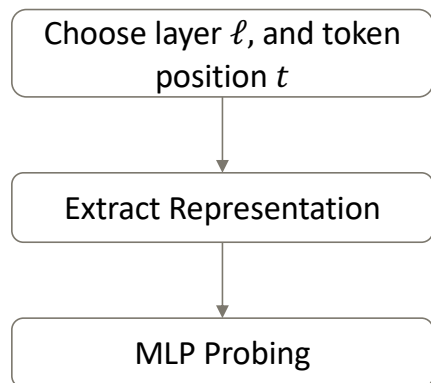
Automatic Layer Selection for Hallucination Detection

Xinpeng Wang¹ · William Cao² · Andrew Gordon Wilson³ · Zhe Zeng¹

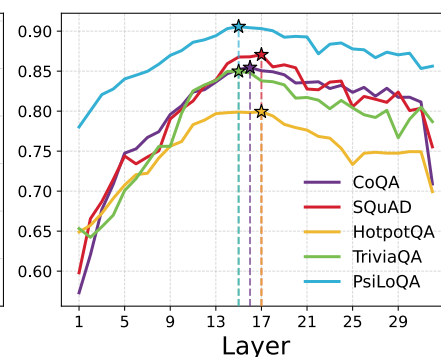
¹University of Virginia · ²Walrus Security · ³New York University

QUESTION 1

Which Layer ℓ Should We Extract?



(a) LLaMA-3.1-8B-Instruct



(b) Mistral-7B-Instruct-v0.3

Best-performing layer consistently lies in the intermediate layers

But **varies** across model and datasets

QUESTION 2

Which Token Position Should We Extract?

Question: Were Scott Derrickson and Ed Wood of the same nationality?

LlaMA-Instruct: No. Scott Derrickson is American, Ed Wood is American but of English descent. However, both are American. So the answer is yes

Mistral-Instruct: No, Scott Derrickson is American and Ed Wood is Canadian.

(a) Inconsistent Continuation

Question: When was the American writer of the a series of four vampire-themed fantasy romance novels born?

LlaMA-Instruct: Anne Rice was born on October 4, 1941. The series is called "The Vampire Chronicles". The first book in the series

Mistral-Instruct: Cassandra Clare was born on July 27, 1971.

(b) Semantic Drift

Question: In what corner of Missouri is radio station WMBH located?

LlaMA-Instruct: The Ozark Mountains. The station is located in the Ozark Mountains of southern Missouri. The station is located in the Ozark Mountains of

Mistral-Instruct: WMBH is located in the southeast corner of Missouri.

(c) Degenerate Repetition

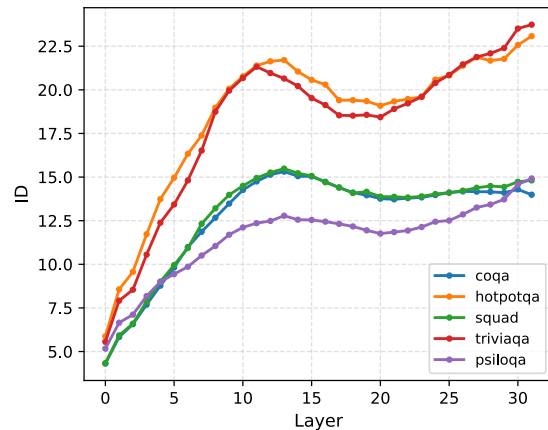
Representation at the last token are always corrupted by end-of-sequence noise

Criteria for Automatic Layer Selection

Hypothesis		Criteria
(i)	Rich semantic information	RankME
(ii)	Task aligned features	Validation Loss
(iii)	Information compression	Curvature
(iv)	High effective information capacity	Intrinsic Dimension (ID); FEPoID

None of existing criteria can efficiently and reliably identify high-performing layers

First Effective Peak of Intrinsic Dimension (FEPoID)



FEPoID

1. Identify all local maximum of $\{d_{\text{ID}}^{(\ell)}\}_{\ell=1}^L$
2. Discard candidate peak at layer ℓ if ID monotonically increase and exceed $d_{\text{ID}}^{(\ell)}$ within a window of size w .
3. Choose the earliest remaining peak

1st peak · abstract semantic information

2nd peak · lexical or surface-level information

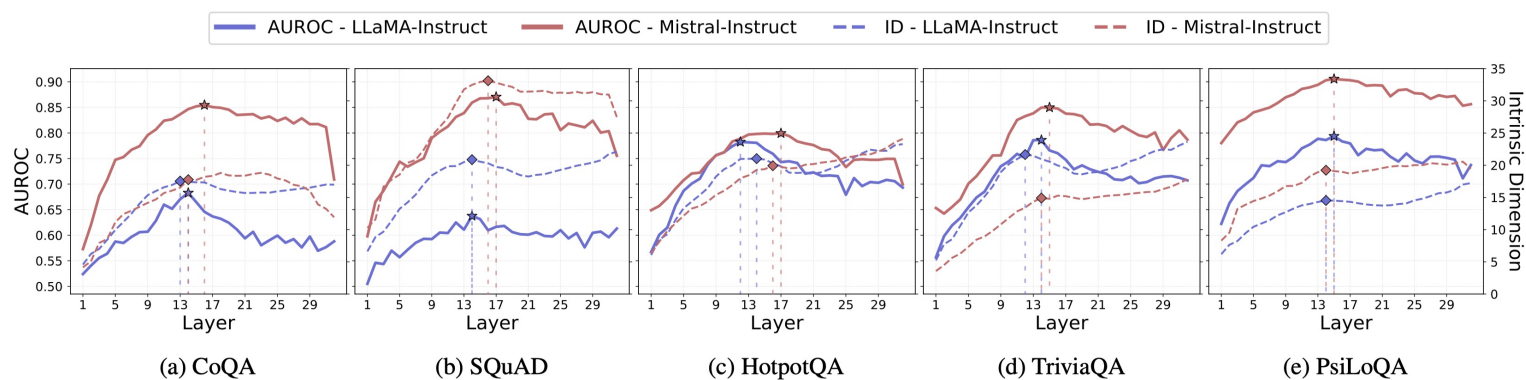
Performance across Models and Tasks

		LlaMA-3.1-8B-Instruct					Mistral-7B-Instruct-v0.3						
		CoQA	SQuAD	HotpotQA	TriviaQA	PsiloQA	Avg	CoQA	SQuAD	HotpotQA	TriviaQA	PsiloQA	Avg
Pred. Entropy		0.5833	0.5703	0.7103	0.6859	0.3604	0.5820	0.7200	0.7316	0.6303	0.6763	0.6110	0.6738
LN-Pred. Entropy		0.5781	0.5671	0.7087	0.6774	0.3555	0.5774	0.6528	0.6390	0.6589	0.6828	0.4418	0.6151
Semantic Entropy		0.5003	0.5518	0.4454	0.5505	0.6076	0.5311	0.5769	0.6409	0.6418	0.7353	0.6853	0.6560
Lexical Similarity		0.6780	0.5988	0.7294	0.6838	0.4082	0.6196	0.7071	0.7169	0.6946	0.7547	0.5781	0.6903
LID		0.5059	0.5281	0.5171	0.4989	0.5994	0.5299	0.5274	0.5688	0.5518	0.4970	0.6447	0.5579
EigenScore		0.5247	0.5300	0.5987	0.5882	0.5080	0.5499	0.7092	0.7508	0.6587	0.7273	0.7246	0.7141
Hidden-State Probing	RankME	0.6598	0.6071	0.7040	0.7114	0.7478	0.6860	0.7743	0.7341	0.6909	0.6699	0.8277	0.7394
	Curvature	0.6323	0.6183	0.7413	0.7366	0.7565	0.6970	0.8492	0.8553	0.7940	0.8368	0.9005	0.8472
	Val Loss	0.6705	0.6164	0.7682	0.7861	0.7836	0.7250	0.8283	0.8679	0.7968	0.8496	0.8861	0.8457
	RGN	0.5993	0.6130	0.7040	0.7859	0.7373	0.6879	0.7090	0.7553	0.7493	0.7868	0.8562	0.7713
	SNR	0.5240	0.5699	0.7009	0.5567	0.6624	0.6028	0.7475	0.6870	0.7620	0.7008	0.8207	0.7436
	ID	0.6705	0.6130	0.6932	0.7073	0.7373	0.6843	0.8283	0.8679	0.6993	0.7868	0.8533	0.8071
	FEPoID	0.6705	0.6377	0.7807	0.7516	0.7862	0.7253	0.8466	0.8679	0.7982	0.8496	0.9031	0.8531

	LlaMA-3.1-8B-Instruct			Mistral-7B-Instruct-v0.3		
	HaluEval	CNN/DM	Avg	HaluEval	CNN/DM	Avg
RankMe	0.6075	0.5774	0.5924	0.7149	0.6869	0.7009
Curvature	0.5494	0.5922	0.5708	0.7498	0.7319	0.7409
Val Loss	0.5961	0.5859	0.5910	0.7294	0.6938	0.7116
RGN	0.5713	0.5821	0.5767	0.7563	0.7031	0.7297
SNR	0.5528	0.5474	0.5501	0.7385	0.6811	0.7098
ID	0.6075	0.5918	0.5997	0.7498	0.7185	0.7342
FEPoID	0.6165	0.5995	0.6080	0.7808	0.7614	0.7711

- Outperforms all layer-selection baselines
- Generalize across QA and summarization tasks

Layer Selection Performance



FEPOID consistently identifies optimal or near-optimal layers

Time Efficiency of FEPoID

	CoQA	SQuAD	HotpotQA	TriviaQA	PsiLoQA	Avg
RankMe	26.32	27.65	28.98	28.17	25.41	27.30
Curvature	43.88	45.23	45.55	46.16	45.37	45.24
Val Loss	27.59	30.81	29.80	29.79	29.78	29.55
RGN	54.83	60.95	58.31	58.08	58.72	58.18
SNR	54.20	59.91	57.31	58.84	59.16	57.88
FEPoID; ID	9.40	9.90	9.95	10.59	10.88	10.14

the **lowest** computation time across all benchmarks

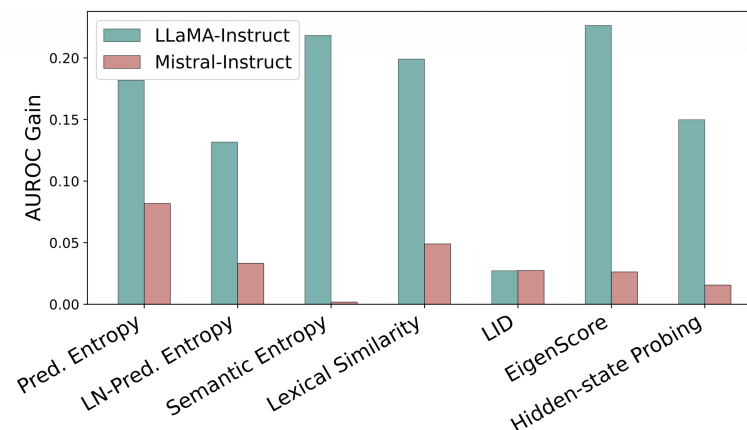
First-Sentence Truncation (FST)

Question: When was the American writer of the a series of four vampire-themed fantasy romance novels **born?** ✕

LlaMA-Instruct: Anne Rice was born on October 4, 1941. The series is called "The Vampire Chronicles". **The first book in the series**

→ Last token heuristic

↓
First sentence truncation



FST substantially improves performance for **all methods** considered

Empirical Analysis of FST

Dataset	Fisher Separation $\uparrow$		Silhouette Score $\uparrow$	
	w/o FST	w/ FST	w/o FST	w/ FST
CoQA	0.0003	0.0044	-0.0007	0.0324
SQuAD	0.0004	0.0049	0.0006	0.0973
HotpotQA	0.0009	0.0075	-0.0001	-0.0011
TriviaQA	0.0013	0.0185	0.0059	0.1006
PsiLoQA	0.0016	0.0340	0.0118	0.2605

Representations extracted with FST exhibit cleaner and more discriminative class structure